THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonal of a rectangle divides it into two: 1

(a) similar squares **(b)** equilateral triangles
(c) congruent right triangles **(d)** isosceles triangles with no right angle

2. The legs of a right triangle are 4 cm and 6 cm. Its hypotenuse (in cm) is: 1

(a) $\sqrt{10}$ **(b)** 10 **(c)** $\sqrt{53}$ **(d)** $2\sqrt{13}$

3. The legs of a right triangle are 8 cm and 17 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 353 **(b)** 625 **(c)** 136 **(d)** 25

4. Taking $m = 11$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 11, 120, 122 **(b)** 22, 120, 123 **(c)** 22, 120, 122 **(d)** 22, 121, 122

5. A triangle has sides 13 cm, 16 cm and 22 cm. It is: 1

(a) cannot be said **(b)** right-angled **(c)** not right-angled **(d)** equilateral

6. The diagonal of a square of side 7 cm (in cm) is: 1

(a) $7\sqrt{2}$ **(b)** 14 **(c)** 7 **(d)** $3\sqrt{11}$

7. The hypotenuse of a right triangle is 37 cm and one leg is 12 cm. Its area (in cm²) is: 1

(a) 420 **(b)** 210 **(c)** 222 **(d)** 211

8. The legs of a right triangle are 1 cm and 7 cm. Its hypotenuse (in cm) is: 1

(a) $\sqrt{51}$ **(b)** $5\sqrt{2}$ **(c)** 8 **(d)** $2\sqrt{2}$

9. The legs of a right triangle are 12 cm and 19 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 228 **(b)** 505 **(c)** 961 **(d)** 31

10. Taking $m = 12$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 12, 143, 145 **(b)** 24, 143, 145 **(c)** 24, 143, 146 **(d)** 24, 144, 145

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the length of the diagonal of a square of side 10 cm. 2

12. Find the Pythagorean triplet with the smallest member 36. 2

13. The sides of a triangle are 19 cm, 21 cm, 29 cm. Is it a right triangle? Which angle is the right angle? 2

14. Find the length of the diagonal of a square of side 12 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The hypotenuse of an isosceles right triangle is $10\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

16. A ladder 61 m long reaches a window 60 m above the ground. Find the distance of the foot of the ladder from the wall. 3

17. The diagonals of a rhombus are 56 cm and 90 cm. Find its side. 3

OR

The hypotenuse of an isosceles right triangle is $12\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

18. A ladder 34 m long reaches a window 30 m above the ground. Find the distance of the foot of the ladder from the wall. 3

OR

The diagonals of a rhombus are 66 cm and 112 cm. Find its side. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The diagonals of a rhombus are 22 cm and 120 cm. Find its side, perimeter and area. 5

OR

Two poles of heights 35 m and 11 m stand on level ground 32 m apart. Find the distance between their tops. 5

20. The diagonals of a rhombus are 32 cm and 60 cm. Find its side, perimeter and area. 5

OR

Two poles of heights 28 m and 4 m stand on level ground 7 m apart. Find the distance between their tops. 5

END OF QUESTION PAPER — Total: 40 Marks

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